

ESG Ratings, Institutional Rebalancing, and the Index Mechanics

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Abstract

This paper studies how institutional investors change their ownership in a firm based on MSCI Environmental, Social, and Governance (ESG) rating changes. I find that institutional ownership rises following rating upgrades and falls when a firm is downgraded below the ESG index-eligibility threshold. Further analysis of the type of institution reveals that dedicated investors drive the ownership response to ESG rating upgrades, whereas transient investors move in the opposite direction, canceling out the ownership decline on downgrades. Further, oftentimes, ESG-focused index rebalancing is confounded by simultaneous changes in the broader parent index. Once I condition on parent-index changes, the ownership decline disappears.

1 Introduction

According to BNP Paribas, approximately 90 percent of institutional investors now incorporate Environmental, Social, and Governance (ESG) considerations into their investment process.¹ And assets in ESG-screened U.S. equity funds have grown from under \$100 billion in 2013 to more than \$500 billion over the last decade. Moreover, MSCI, a key intermediary in the ESG ecosystem, has roughly 70 percent of the global ESG rating market and more than \$1 trillion in assets benchmarked to its sustainability indices.² As a result, ESG ratings are important in that they shape institutional portfolio construction and determine a stock’s eligibility for MSCI ESG indices.

This paper examines the link between ESG ratings and institutional ownership, with emphasis on whether or not ESG ratings changes lead institutional investors to rebalance their portfolios. I also examine the channels that drive ownership changes and the extent of variation in such changes across investor types.

I examine these questions using a sample of U.S. publicly traded firms from 2013 to 2023, combining MSCI ESG ratings, SEC Form 13F institutional ownership data, and index constituent records from ESG-focused Equity Traded Funds. I use three approaches for identification. The rating change itself, MSCI’s BB index-eligibility threshold (discussed further in detail below), and ESG ETF rebalancing.

The first analysis asks whether the ratings, separate from any index or ETF considerations, shift institutional ownership by treating MSCI ESG rating upgrades and downgrades as quasi-exogenous shocks to perceived ESG quality. I then estimate the effect of such ratings changes on quarterly 13F holdings using two-way fixed effects, propensity score matching (PSM), and difference-in-differences estimation. The results show that upgrades result in meaningful increases in institutional ownership. In contrast, aggregate downgrades are not statistically significant in the baseline fixed-effects analysis, but PSM-matched estimates show ownership declines for downgraded firms relative to

¹See [the BNP Survey](#) and [summary article](#) here.

²See [MSCI Report July 2025](#)

matched controls. Post-event ownership declines persist for at least four quarters, and pre-event trends are flat, supporting a causal interpretation. Additional robustness specifications corroborate these patterns.

At the same time, an investor-type decomposition following Bushee (2001) shows that the ownership response differs across the three investor classes. For upgrades, Dedicated (DED) investors increase their holdings, Quasi-index (QIX) investors show no meaningful response, and Transient (TRA) investors reduce holdings. For downgrades, both DED and QIX reduce holdings, while TRA investors move in the opposite direction.

The second analysis focuses on the BB threshold in MSCI’s rating system, which serves as the eligibility criterion for inclusion in ESG-linked indices and thus ETFs tracking them. Note that being upgraded across the threshold is a necessary but not sufficient criterion for index inclusion, while being downgraded below the threshold makes a firm ineligible for inclusion in the index.

Thus, all else equal, this is a shock to the perceived quality of a firm’s ESG, which in turn may trigger ownership changes. That is, I hypothesize that firms crossing the BB/B boundary should experience discrete shifts in ownership as ESG-mandate institutions reassess the quality of a firm’s ESG and/or the likelihood of their addition/deletion from ESG indices. I test this using a regression discontinuity design at the BB/B boundary and a DiD comparing BB-threshold downgrades (from BB to B) with firms experiencing no rating change over the same window. The RDD analysis does not show a statistically significant discontinuity in institutional ownership at the BB threshold. This is consistent with Berg, Cornaggia, et al. (2026), who document a traffic-light effect and that MSCI actively manages threshold crossings. The DiD estimates, however, indicate that BB-threshold downgrades are associated with larger declines in institutional ownership. Taken together, these results suggest that BB-threshold downgrades carry economically meaningful ownership consequences, but I find no evidence that the effect reflects a statistically significant discontinuity precisely at the threshold.

The third analysis studies the index inclusion channel. Barontini and Gioja (2025)

show that ESG index additions and removals generate ownership responses. However, they do not address the question of whether the BB threshold itself is the trigger. To proxy for index additions and deletions, I focus on firms being added to, or dropped from, the portfolios of 39 MSCI ESG-focused ETFs.³ I then estimate the effect of these index changes on institutional ownership alongside the rating change indicators. In the baseline specification, the rating upgrade coefficient is both positive and significant. Additions show no significant effect in this same specification. ESG ETF removals are associated with a decrease in institutional ownership by 1.361 percentage points. These effects are broadly comparable to Barontini and Gioja (2025), who find negative ownership effects at ESG index deletions globally.

At the same time, however, discerning any ESG-index addition effect is challenging in those where MSCI conditions ESG-focused index membership on concurrent membership in a broader MSCI parent index. For example, under MSCI’s methodology, a security must be included in the MSCI USA index before it can appear in any MSCI ESG index. As a result, if a firm is added or removed from an ESG ETF contemporaneously with a change in the parent broad-market index, any resulting change in market value or ownership captures both ESG index and broader index effects. I call these bundled events.

To examine this issue, I map additions/deletions using the ETF holdings for each ESG ETF to its underlying parent broad-market index and flag events where the ETF event and the parent-index event happen in the same calendar month. For additions, this is rare, with only 420 of 16,350 events (2.57 percent) being bundled. For deletions, it is far more common, with 1,694 of 8,988 events (18.85 percent).

The rating upgrade coefficient of 0.278 stays positive and significant throughout, regardless of conditioning on the parent-index changes. Further, the ownership effect remains insignificant for the additions. This is consistent with markets anticipating these events early, meaning the impact is already fully priced in and undetectable in

³I proxy index membership with ETF holdings because MSCI’s ESG index constituent data is not publicly available and is costly to acquire. The daily-disclosed ETF holdings closely track the underlying index membership.

quarterly 13F data at execution (Greenwood and Sammon (2025)). Once bundled events are excluded, the removal coefficient becomes statistically insignificant, indicating that the deletion effect is largely driven by broad-market indexes rather than ESG factors.

Berg, Cornaggia, et al. (2026) document that MSCI algorithmically manages threshold crossings. Firms that fall below BB are not automatically removed from MSCI's ESG indices because additional governance screens smooth the transition. The absence of an ownership discontinuity at the BB/B boundary is fully consistent with their finding and extends it to the ownership channel. The larger ownership effect at BB downgrades is more consistent with investors treating a fall below the eligibility threshold as a signal of deterioration in ESG quality, regardless of whether index removal follows, rather than mechanically responding to the threshold crossing itself.

Beyond these three channels, I also document a coverage-initiation effect. I find that firms receiving their first MSCI ESG rating experience a 1.16 percentage point increase in institutional ownership over the two quarters following coverage initiation, identified by the staggered timing of MSCI's coverage expansion. The visibility effect is consistent with ESG-mandate investors being unable to hold firms until MSCI assigns a rating, so that initiation itself relaxes a holdings constraint regardless of the rating level assigned.

Finally, I examine whether the rating changes are accompanied by price effects. ESG ratings changes generate little detectable abnormal return around the rating change day. For upgrades, the event-day CAR is not significant, whereas for downgrades, it is positive and small. Further, consistent with Greenwood and Sammon (2025), the ESG-focused ETF additions show no robust valuation effect, and ESG-focused ETF removals are accompanied by significant negative abnormal returns. The coverage initiation is associated with a positive 3-day CAR, consistent with a demand-driven price effect at the time of first MSCI coverage.

This paper makes four contributions to the literature. First, I show that institutional investors' portfolio rebalancing appears to track MSCI ratings changes rather than index

membership. Once the ratings channel and the index-addition channel are separated, the rating signal is what moves ownership.

Second, I trace this result to an institutional feature of how MSCI's ESG indices are constructed. Because ESG index membership requires concurrent membership in a broader parent index, ESG index adjustments are frequently bundled with parent-index changes. In my sample, 18.85 percent of deletion events and 2.57 percent of additions. Isolating this bundling requires mapping each ESG index membership to its parent broad-market index, a step the ownership literature has not previously taken. Once I do so, the ETF deletion coefficient becomes statistically indistinguishable from zero, while the rating-upgrade coefficient is positive and significant throughout. Barontini and Gioja (2025) document significant ownership effects at global ESG index deletions; my results suggest the parent-index structure documented here is a natural next step for identifying ESG-specific demand in that setting, and more broadly for any design that uses MSCI ESG index events to proxy for ESG-mandate rebalancing.

Third, the BB-threshold result adds a methodological nuance to the ESG-rating literature. Firms downgraded below BB see larger ownership declines, suggesting investors treat the eligibility threshold as informative even though the RDD shows no sharp ownership discontinuity exactly at the BB/B boundary. Berg, Cornaggia, et al. (2026) help explain why: MSCI actively manages threshold crossings through additional governance screens rather than triggering automatic removals, so a discrete ownership break at the boundary is not expected.

Fourth, decomposing the ownership response by investor type reveals meaningful heterogeneity. For upgrades, the increase in holdings is driven by Dedicated (DED) investors, the long-horizon active holders most likely to respond to improvements in ESG quality; Quasi-Indexed investors show little response. For downgrades, both Dedicated and Quasi-Indexed investors reduce holdings, while Transient investors move in the opposite direction, consistent with liquidity provision. This heterogeneity helps identify who is actually responding to rating changes, and suggests the mechanism differs between

upgrades and downgrades.

The remainder of the paper proceeds as follows. Section 2 reviews related literature and develops hypotheses. Section 3 describes the institutional setting and data. Section 4 explains sample construction and variable definitions. Section 5 outlines the empirical strategy. Section 6 presents the results, including robustness and supplemental analyses. Section 7 concludes.

2 Literature Review and Hypothesis Development

ESG considerations play a central role in investment mandates, and sustainability rankings measurably shift fund flows (Hartzmark and Sussman (2019), Gillan, Koch, and Starks (2021)). Long-term institutions tilt toward high-ESG firms and exit following negative ESG incidents (Starks, Venkat, and Zhu (2017)), and Dyck et al. (2019) find that non-U.S. foreign institutional investors drive ESG improvements through their ownership stakes. Contrary to this, Raghunandan and Rajgopal (2022) find that ESG funds hold firms with worse actual ESG outcomes despite higher reported scores, while Rzeźnik, Hanley, and Pelizzon (2022) find that ownership responses to rating changes are transitory and driven primarily by retail investors. A key source of this ambiguity is rating disagreement: unlike credit ratings, ESG ratings from different providers correlate weakly. Berg, Kölbel, and Rigobon (2022) document a cross-provider correlation of approximately 0.3, and greater disclosure amplifies rather than resolves disagreement (Christensen, Serafeim, and Sikochi (2022)). Rating uncertainty weakens the institutional demand channel (Avramov et al. (2022)).

Isolating how institutions react to ESG rating changes is difficult because these changes trigger multiple effects simultaneously. They provide new information, affect index-eligibility thresholds, and force ESG-focused ETF rebalancing. Because institutions differ systematically in horizon and trading style, their responses to ESG signals need not be uniform. Bushee (2001) classifies institutions as dedicated, quasi-indexed, and

transient: long-horizon dedicated holders are the most plausible carriers of explicit ESG mandates, quasi-indexers track benchmarks mechanically, and transient investors trade on short-term signals and may provide liquidity when others rebalance. Large shareholders can also discipline management through exit threats or voice (Admati and Pfleiderer (2009); Edmans (2009)), and active engagement by institutional shareholders can enhance governance outcomes, including sustainability initiatives (Dimson, Karakas, and Li (2015)). Competitive pressures among fund managers, however, may dilute the effectiveness of exit as a governance tool (Dasgupta and Piacentino (2015)).

To address these identification challenges, I focus on MSCI ratings, the dominant ESG data provider with roughly a 70% market share and over \$1 trillion in assets benchmarked to its Sustainability and Climate Indexes — more than four times its nearest competitor. A distinct strand of the literature isolates ESG-specific demand using index and ETF membership events, and this paper relates most closely to that work. Barontini and Gioja (2025) use official MSCI press releases to identify additions to and deletions from eleven global ESG indices over 2011–2021, documenting that short-term investors reduce holdings by roughly 2.48 percentage points per month around deletions. I take a different approach by investigating the underlying eligibility mechanism. A firm’s presence in an ESG sub-index is explicitly tied to its baseline parent broad-market index membership, so an ESG rebalancing that coincides with a parent-index addition/removal could in part reflect broad-market rebalancing rather than ESG-related effect. Conditioning on parent-index membership is therefore necessary to isolate the ESG-specific component.

It is also unclear whether the eligibility threshold serves as a sharp, mechanical trigger. Berg, Cornaggia, et al. (2026) find that MSCI manages crossings near the BB/B boundary algorithmically — firms that fall below BB are not automatically dropped from ESG indices, as additional governance screens smooth the transition. Because this smoothing reduces variation right at the cutoff, regression-discontinuity tests around the threshold tend to have limited power, which is one reason this paper relies mainly on difference-in-differences and propensity-score matching.

In this paper, I propose two core hypotheses:

Hypothesis 1 (Institutional Ownership). *Null* ($H_{0,1}$): ESG rating changes have no effect on institutional ownership. *Alternative* ($H_{A,1}$): ESG rating upgrades increase institutional ownership, while downgrades decrease it, with the response driven by the informational content of the rating rather than mechanical index inclusion, and concentrated among long-horizon dedicated investors.

Hypothesis 2 (Firm Value). *Null* ($H_{0,2}$): ESG rating changes have no effect on firm value as reflected in stock prices. *Alternative* ($H_{A,2}$): ESG rating changes and first-time MSCI coverage initiation trigger abnormal stock-market reactions.

These hypotheses build on and connect several strands of this literature. The hypothesis that ownership responds to rating changes follows from work showing that ESG preferences shape institutional portfolios (Dyck et al. (2019); Raghunandan and Rajgopal (2022); Agrawal et al. (2023)). The idea that downgrades matter more than upgrades fits the governance literature on exit and voice (Admati and Pfleiderer (2009); Edmans (2009); Appel, Gormley, and Keim (2016)) and models in which investors monitor selectively, responding when thresholds are crossed or reputational risks rise (Broccardo, Hart, and Zingales (2022); Gollier and Pouget (2014); Pedersen, Fitzgibbons, and Pomorski (2021)).

3 Institutional Setting

3.1 ESG Ratings

Environmental, Social, and Governance (ESG) ratings are produced by several independent agencies, including MSCI, Moody's, Sustainalytics, ISS, and S&P among others. While these providers aim to assess corporate sustainability, there is considerable variation in methodologies and significant disagreement in firm-level ESG ratings across agencies (Christensen, Serafeim, and Sikochi, (2022); Avramov et al., (2022); Berg, Kölbel, and

Rigobon, 2022). In this study, I focus exclusively on MSCI ESG ratings, which are among the most widely adopted in the U.S. investment landscape and have been shown to exhibit the strongest alignment with the portfolio decisions of ESG-focused funds (Berg, Kölbel, and Rigobon, 2022).⁴

MSCI derives ESG ratings from a combination of structured data sources, including regulatory filings, company disclosures, media reports, NGO assessments, and other stakeholder inputs. These inputs inform scores across multiple environmental, social, and governance key issues. The overall ESG rating is calculated as a weighted average of pillar-level scores, adjusted for relative comparison with industry peers. Firms are then assigned a rating on a seven-point ordinal scale, ranging from AAA (leader) to CCC (laggard), with adjustments designed to account for sector-specific risks and opportunities.⁵

To simplify interpretation, MSCI further classifies firms into three rating tiers: Leaders (AAA, AA), Average (A, BBB, BB), and Laggards (B, CCC). While most ESG ratings are updated annually, MSCI incorporates interim data revisions—such as controversies or material disclosures—as inputs to rating adjustments throughout the year. For many ESG-oriented funds, MSCI ratings directly inform portfolio construction, both through explicit mandates and index inclusion rules. For instance, many ESG indices require firms to maintain a rating above BB to remain eligible, and ETFs following these indices reflect the same eligibility constraints. Accordingly, I place emphasis on rating changes that cross the BB-to-BBB threshold, as such changes have implications for index eligibility and investor flows.

This focus not only reflects MSCI’s influence in shaping ESG-driven investment strategies but also facilitates the identification of quasi-exogenous shocks to firms’ perceived ESG quality. These shocks allow for a cleaner empirical design when testing how investors and markets respond to changes in ESG assessments.

MSCI also expanded its ESG coverage in waves, with a notable broadening in

⁴See Opimas (2022): “ESG Data Is Now Worth It,” <https://www.opimas.com/research/742/detail/>

⁵See MSCI ESG Methodology: <https://www.msci.com/sustainable-investing/esg-ratings>.

2012Q4 in which 1,602 previously uncovered U.S. firms received their first MSCI ESG rating in the same quarter. Because the timing of this expansion was determined by MSCI’s product roll-out rather than by firm-level decisions, it provides a useful setting for identifying the effect of initial ESG visibility on institutional ownership and market reactions, which I exploit in Section [6](#).

3.2 Institutional Ownership

To examine how institutional investors respond to ESG rating changes, I use institutional holdings data compiled from SEC Form 13F filings, accessed via the Wharton Research Data Services (WRDS) platform. Under SEC regulations, institutional investment managers with at least \$100 million in assets under management are required to file Form 13F each quarter.

Each filing includes information on the number of shares held, issuer names, ticker symbols, and market value, enabling comprehensive tracking of institutional equity positions over time. The 13F dataset captures a broad spectrum of institutional investors, ranging from asset managers and pension funds to hedge funds and insurance firms, offering a rich view of investor behavior in response to ESG-related information.

For this study, I track changes in institutional ownership around ESG rating events, with a particular focus on MSCI ratings upgrades and downgrades. By analyzing these changes, I aim to determine whether ESG signals result in measurable capital reallocation by institutions. This is particularly relevant for understanding whether investors use ESG information as part of an “exit” mechanism, consistent with governance theories that emphasize shareholder discipline (Admati and Pfleiderer, [2009](#); Edmans, [2009](#)) and whether such exits lead to measurable changes in broader institutional holdings (Berk and Van Binsbergen, [2025](#)).

In addition to broad institutional holdings, a distinct channel operates through ESG-focused indices. These indices use MSCI ESG Ratings to identify companies that

effectively manage ESG risks and opportunities, and firms must maintain a rating of BB or higher to be eligible for inclusion in the indexes⁶—so rating changes can trigger additions to or removals from ESG-oriented indices. Due to data limitations on index-level additions and deletions, I proxy this mechanism using changes in ETF holdings. For example, the iShares ESG Optimized MSCI USA ETF⁷ explicitly seeks to track its ESG sub-index (MSCI USA Extended ESG Select Index) composed of U.S. companies with positive ESG characteristics, as identified by MSCI. Since ESG ETFs adjust their portfolios to mirror ESG sub-index revisions, these changes serve as a useful proxy for identifying rating-based shifts in index constituents and, in turn, institutional ownership.

Further, ESG sub-indices adjust their constituents based on both ESG rating changes and changes in the parent broad-market index. To isolate the effect of rating changes on institutional ownership, I identify and exclude events in which ESG index additions or deletions coincide with concurrent changes in parent broad-market index membership.

4 Sample Construction

Although MSCI provides ESG ratings beginning in 2007, comprehensive coverage became available in 2012, and consistent institutional holdings data from SEC Form 13F are available starting in 2013. Accordingly, the primary sample spans January 2013 through December 2023. I obtain ESG ratings from MSCI, merge them with institutional holdings from WRDS 13F filings, and supplement with firm-level controls from the CRSP–COMPUSTAT merged database. After combining these sources, the final sample comprises 28,571 firm-year observations, averaging roughly 2,600 firms per year.

[INSERT FIGURE 1 HERE]

Figure 1 illustrates the distribution of ESG ratings across the sample. Firms cluster

⁶See MSCI Selection Indexes Methodology (page 4).

⁷See ETF fact sheet.

heavily in the “Average” category, with nearly 70% of observations rated either BB or BBB. This distribution is notable because many ESG indices apply BB as a minimum eligibility cutoff, making firms at or near this threshold especially relevant for index inclusion decisions.

To analyze how institutions respond to these ESG rating shocks, I construct a quarterly panel by merging three data sources: MSCI ESG ratings (Final Industry-Adjusted Score), institutional ownership from WRDS SEC 13F filings, and firm financials from the CRSP–COMPUSTAT Merged (CCM) database. Because MSCI reports ESG ratings monthly while 13F filings are quarterly, I aggregate ratings to the quarterly frequency. The resulting dataset consists of 105,086 firm-quarter observations. The primary dependent variable is institutional ownership, measured as the percentage of shares held by 13F filers.

[INSERT TABLE 1 HERE]

Table 1 reports summary statistics for the firm-quarter-level sample. Institutional ownership averages 71.7% of shares outstanding, with a median of 79.2%, reflecting the significant institutional presence in these firms. ESG ratings are mapped from MSCI’s seven-point scale, where 1 represents CCC and 7 represents AAA. The average rating is 3.54, corresponding roughly to a BB/BBB classification. Control variables include leverage, firm size (log market capitalization), market-to-book ratio, profitability, and quarterly stock returns.

[INSERT TABLE 2 HERE]

Rating changes over the sample period are summarized in Table 2. The transition matrix highlights two features: first, the diagonal values are large, indicating that most firms retain their prior rating from quarter to quarter; second, meaningful movement occurs between adjacent categories, particularly between A and BBB, and between BB and B. This moderate rating volatility, concentrated around the BB threshold, has direct

implications for ESG index eligibility. I identify 4,539 upgrades and 2,710 downgrades, with 1,233 upgrades and 1,113 downgrades occurring specifically at the BB threshold. These “rating change events” represent quasi-exogenous shocks to firms’ perceived ESG status and serve as the key identification strategy in my empirical design.

In addition to rating changes, I identify ETF additions and removals from the constituent holdings of ESG-focused ETFs reported in WRDS. Each ETF event is aligned to the corresponding rating change window for use in the multivariate specification described in Section 5.

For the coverage-initiation analysis, I construct a separate sub-sample that extends back to the period preceding MSCI’s 2012Q4 coverage expansion. The sub-sample covers 2009–2014, combining monthly MSCI ESG ratings with quarterly 13F institutional holdings, merged by CUSIP and calendar quarter. It includes the 1,602 firms that received their first MSCI ESG rating in 2012Q4, together with control firms that were already rated, observed over event quarters $[-2, +2]$ around initiation. Outcomes for this sub-sample are institutional ownership from 13F filings and short-window cumulative abnormal returns estimated using the market-adjusted and Fama–French three-factor models, with a 252-day estimation window and a 30-day gap before the event. Additional details are provided in Section 5.

A potential confound in identifying ESG-specific institutional demand is that major ESG index providers condition sub-index membership on concurrent membership in a parent broad-market index. A firm entering an ESG ETF and its underlying parent index in the same calendar month is subject to two simultaneous demand shocks, the ESG-specific rebalancing and the standard index-inclusion effect. This could overstate inferences about ESG-driven ownership changes. To address this, I map each of the 39 ESG-focused ETFs in the sample to its underlying parent broad-market index and identify the corresponding proxy ETF for that parent. For each ESG ETF addition event, I flag the event as bundled if the firm first appears in the matched parent ETF in the same calendar month. I apply the same criterion symmetrically to deletions: a deletion is bundled if

the firm exits the matched parent ETF in the same calendar month. Bundled additions are rare, comprising 498 of 16,383 ESG ETF addition events (3.04 percent). Bundled deletions are substantially more common, comprising 1,694 of 8,989 ESG ETF deletion events (18.85 percent). In the main analysis, I construct Clean Added and Clean Removed indicators that equal zero for bundled events, isolating ownership changes attributable to ESG rating signals from those driven by concurrent parent-index rebalancing. The details about full parent-index mapping and the incidence of bundled events by index family is available in the online appendix.

5 Methodology

5.1 Univariate Analysis

I begin with a univariate analysis to explore the association between ESG ratings and ownership structure, proxied by institutional holdings reported in 13F filings. Specifically, I first examine how institutional ownership varies across the seven ESG rating levels. To test for overall differences, I conduct an F-test (ANOVA) across all rating groups, and to assess threshold effects more directly, I complement this with an independent samples t-test comparing firms above and below the BB cutoff. This dual approach provides descriptive evidence on whether ownership patterns vary systematically across the ESG spectrum and whether the BB threshold in particular marks a discontinuity in institutional investment behavior.

To assess how financial markets respond to ESG rating changes, I conduct a standard event study analysis. I calculate cumulative abnormal returns (CAR) using a market model, where abnormal returns are computed as the deviation from expected returns over multiple event windows: short-term windows $[0, 0]$, $[0, 2]$, $[0, 4]$ and extended windows $[-1, 0]$, $[-2, 2]$, $[-4, 4]$ to capture potential information leakage. I then compare mean CARs across upgrade and downgrade events, as well as between BB-threshold and non-threshold changes. This expanded univariate framework provides initial descriptive

evidence of how both ownership and market value respond to changes in ESG ratings.

5.2 Addressing Endogeneity

While informative, the univariate analysis is limited by potential endogeneity concerns, including reverse causality and omitted variable bias. For instance, it is difficult to establish whether institutional investors influence ESG outcomes or respond to them. Moreover, unobserved factors may simultaneously affect both ownership structure and ESG ratings, creating bias in simple comparisons.

To address these issues, I use changes in ESG ratings as a quasi-exogenous shock to firms' ESG profiles. These rating changes are plausibly outside direct managerial control and provide a setting resembling a natural experiment. I then assess the effects of these rating shocks on institutional ownership and firm value using a two-way fixed effects (FE) regression model that accounts for unobservable heterogeneity at both the firm and time levels.

The following specification is estimated:

$$13F\ Holding_{i,t+1} = \alpha + \delta Upgrade_{i,t} + \gamma Downgrade_{i,t} + \rho BBUpgrade_{i,t} + \sigma BBDowngrade_{i,t} + \mu_i + \theta_t + \epsilon_{i,t} \quad (1)$$

The variables *Upgrade* and *Downgrade* are indicator variables equal to 1 if there is an upgrade or downgrade in ESG rating for the firm in quarter t , and 0 otherwise. *BBUpgrade* and *BBDowngrade* are also indicator variables equal to 1 if the upgrade or downgrade crosses the BB threshold. The terms μ_i and θ_t capture firm and time fixed effects, respectively.

5.3 Addressing Selection Bias

To further mitigate bias arising from non-random treatment assignment, I implement a Propensity Score Matching (PSM) procedure. Selection bias can arise if, for example,

larger firms simultaneously improve ESG performance and attract institutional investors. Similarly, institutional investors may systematically favor certain types of firms—such as those in specific industries or with particular ESG profiles.

To correct for this, each treated firm (i.e., those experiencing a rating change) is matched with a control firm that did not experience a rating change in the same quarter, based on a propensity score estimated from firm size (log of market capitalization), industry (as defined by MSCI), and prior ESG rating. Matching is performed separately for upgrades and downgrades.⁸

After matching, I compute the Average Treatment Effect (ATE) and the Average Treatment Effect on the Treated (ATET) for institutional holdings. This approach reduces the influence of observable confounders and strengthens causal interpretation.

5.4 Difference-in-Differences Approach

To complement the PSM approach and allow for pre- and post-event comparisons, I implement a Difference-in-Differences (DiD) framework. This method compares the evolution of institutional holdings around ESG rating changes (treatment firms) with that of matched firms without rating changes (control firms).

The DiD regression model is specified as follows:

$$13F\ Holding_{i,t} = \beta d[s + k]_{i,t} + \gamma RatingChange \times d[s + k]_{i,t} + \mu_i + \theta_t + \epsilon_{i,t} \quad (2)$$

where

- $d[s + k]$ with $-2 \leq k \leq 2$ is a set of indicator variables for the quarter relative to a rating change or placebo quarter, with quarter $s - 1$ as the omitted base quarter.

⁸I also explored adding pre-event institutional ownership as an additional matching criterion. The algorithm, however, cannot simultaneously match on both ownership level and pre-event ESG rating within caliper = 0.05: adding ownership to the matching set causes the pre-ESG rating to lose balance (standardized bias of 25.8%; overall balance statistic $B = 28.7$, $p < 0.001$). In the ownership-augmented specification, the BB downgrade treatment effect in the firm and quarter fixed effects regression shrinks to near zero (-0.04 , $p = 0.932$).

Thus, $d[s + k]$ captures the difference in the level of 13F holdings among control firms between quarters $s - 1$ and $s + k$.

- The interaction term $\text{RatingChange} \times d[s + k]$ captures the difference in 13F holdings between $s - 1$ and $s + k$ between firms experiencing rating changes and control firms.
- μ_i represents firm fixed effects, controlling for time-invariant firm characteristics.
- θ_t represents time fixed effects, controlling for time-specific factors common to all firms.

As a robustness check, I also implement local-projection difference-in-differences (Dube et al., 2023), which accommodates dynamic and staggered treatment effects more flexibly than the standard DiD specification. The estimator traces the response of 13F holdings at successive event horizons relative to the rating change, with standard errors clustered at the firm level. Full estimation results are reported in online Appendix.

5.5 Investor-Type Heterogeneity

To examine whether the ownership response to ESG rating changes is concentrated among particular investor types, I decompose institutional holdings into the Bushee (2001) classification: Dedicated (DED), Quasi-Indexer (QIX), and Transient (TRA). Dedicated investors hold concentrated, long-horizon positions and are the most plausible carriers of explicit ESG mandates. Quasi-indexers track benchmarks mechanically. Transient investors trade on short-term signals. I re-estimate the baseline TWFE specification in equation (1) separately for each investor class as the dependent variable, and additionally include interactions between the rating change indicators and investor-type dummies in a pooled specification. This decomposition tests whether the aggregate ownership response masks heterogeneous behavior across investor types and whether the ESG mandate channel operates primarily through long-horizon investors.

5.6 ETF Mechanism: Additions and Removals

To more cleanly examine the index-linked channel through which ESG ratings affect institutional ownership, I study additions to and removals from ESG-focused ETFs. Because ESG indices typically require firms to maintain a rating of BB or higher as an eligibility cutoff, rating changes can trigger inclusion in or exclusion from ESG indices—and, consequently, from the ETFs that track them. Since comprehensive index-level rebalancing data are not available, I use firms added to or removed from ESG-focused ETFs as a proxy for this mechanism.

A complication arises because ESG sub-index membership is conditioned on concurrent membership in a parent broad-market index. A firm entering an ESG ETF and its parent index in the same calendar month is subject to two simultaneous demand shocks, and the raw Added and Removed indicators conflate ESG-specific rebalancing with the standard index-inclusion effect. To isolate the ESG channel, I map each ESG ETF to its underlying parent broad-market index and identify a proxy ETF for that parent. For each ETF addition and deletion event, I flag the event as bundled if the firm also enters or exits the matched parent ETF in the same calendar month. I then construct Clean Added and Clean Removed indicators that equal zero for bundled events. Bundled additions comprise 498 of 16,383 ESG ETF addition events (3.04 percent of additions); bundled deletions are substantially more common, comprising 1,694 of 8,989 deletion events (18.85 percent).

I extend the baseline 13F ownership specification by including indicators for ETF additions and removals, alongside ESG rating changes. This regression framework tests whether institutional ownership responses to rating changes are stronger when firms are added to or removed from ESG ETFs, providing a cleaner proxy for underlying index additions and deletions.

I also conduct an event study of cumulative abnormal returns (CARs) around rating changes and ETF events. I define an ETF event as the month in which a firm is

first added to or dropped from the holdings of an ESG-focused ETF around an MSCI rating change. These events are identified from ETF constituent data through WRDS and are aligned relative to the timing of the corresponding rating revision. Abnormal returns are computed using a market model over multiple windows ($[0, 0]$, $[0, 2]$, $[0, 4]$), with additional horizons ($[-1, 0]$, $[-2, 2]$, $[-4, 4]$) to capture potential information leakage. I distinguish four timing conditions: the full sample, ETF additions and removals in the same month as the rating change, those within four months, and those occurring beforehand.

Together, this combined regression and event study approach isolates whether changes in ownership and market value reflect the informational effects of ESG ratings.

5.7 Coverage Initiation as a Quasi-Natural Experiment

As a quasi-natural experiment that further mitigates endogeneity concerns, I exploit MSCI’s mass coverage initiation in the fourth quarter of 2012, when 1,602 previously uncovered firms received their first MSCI ESG rating in the same quarter. The timing of this expansion was determined by MSCI’s product roll-out rather than by firm-level decisions, making the assignment of treatment plausibly exogenous to firm fundamentals. This setting allows me to assess how the initial appearance of an ESG rating—rather than a change in an existing rating—affects institutional ownership and stock returns.

For each newly rated firm, event quarter s is defined as the quarter of initial coverage. I track institutional ownership and cumulative abnormal returns over an event window of $[-2, +2]$ quarters around initiation, using firms that were already rated as a comparison group. For ownership, I compare mean 13F holdings across event quarters and test for differences using paired and two-sample t-tests. For market reactions, I estimate cumulative abnormal returns using both the market-adjusted model and the Fama–French three-factor model, with a 252-day estimation window and a 30-day gap before initiation. CARs are computed over short-term windows ($[0, 0]$, $[0, 2]$, $[0, 4]$).

To examine heterogeneity, I split the sample by the firm’s initial rating at the time of coverage initiation (AA, A, BBB, BB, B, and CCC, with AAA omitted due to insufficient observations). This decomposition reveals whether market reactions are concentrated in firms whose first rating conveys particularly informative signals.

6 Results

6.1 Univariate Evidence

I begin with a univariate analysis to examine whether institutional ownership varies systematically across ESG rating categories. Panel A of Table 3 shows a clear positive association between ESG ratings and 13F institutional holdings. Mean holdings rise with ESG rating, from 61.89 percent for CCC-rated firms to 78.64 percent for AA-rated firms. The median ownership follows a similar pattern. An ANOVA F-test rejects the null of equal means across rating categories. Economically, the pattern implies that higher-rated firms have substantially greater institutional ownership.

[INSERT TABLE 3 HERE]

Panel B splits the sample at the BB eligibility threshold. Firms rated BB and above had an average institutional ownership of 73.28 percent, compared with 66.28 percent for firms rated below BB. The difference of approximately 7 percentage points is statistically significant. Panel C focuses on firms that cross the BB boundary through a rating change. Firms upgraded above the BB threshold average 70.38% institutional ownership, while those downgraded below average 67.09%, a statistically significant difference of 3.28 percentage points. Taken together, these univariate results suggest that institutional ownership is positively associated with ESG standing and is particularly elevated around the BB threshold.

These patterns are descriptive and do not control for concurrent changes in firm

characteristics or market conditions. Higher-rated firms tend to be larger, more profitable, and more liquid than lower-rated firms, as Panel A also shows through the monotonic increase in log market capitalization across rating categories. It is therefore possible that the positive ownership-rating relation is partly a size premium in institutional demand rather than ESG-specific allocation. The fixed-effects and matching specifications in the following subsections address this concern by conditioning on firm characteristics and by comparing firms to themselves over time.

6.2 Rating Signal and Institutional Ownership

To control for stable firm characteristics and common time trends, I estimate two-way fixed effects regressions with firm and quarter fixed effects. The dependent variable is the percentage of outstanding shares held by 13F-filing institutions in each calendar quarter. The key regressors are indicator variables set to one in the quarter of an ESG upgrade or downgrade, and separate indicator variables that restrict attention to rating changes that cross the BB eligibility threshold governing inclusion in major ESG indices. All specifications include firm and quarter fixed effects and a vector of firm-level controls, including leverage, log market capitalization, market-to-book, profitability, and quarterly stock return.

Table 4 reports the results across three specifications. Specification (1) includes only the general upgrade and downgrade indicators. Upgrades are associated with a statistically significant 0.203% increase in institutional ownership. Downgrades have a coefficient of -0.142, which is not statistically significant. Specification (2) examines the effect of rating changes that cross the BB threshold. The BB downgrade indicator is -0.491, negative in sign and directionally consistent with ESG-index-driven selling, though not statistically significant. Specification (3) includes all four indicators jointly. The upgrade coefficient expands to 0.278% and is significant. The BB downgrade coefficient is -0.526.

[INSERT TABLE 4 HERE]

The upgrade effect is more precisely estimated than the downgrade effect, though a remaining concern is that firms receiving upgrades may already be attracting institutional demand prior to the rating change. I address this in the next section.

6.3 Propensity Score Matching and Dynamic DiD

To address selection bias, I construct matched control samples using propensity score matching (PSM). For each treated firm, I identify the nearest-neighbor control firm from the set of firms that did not experience a rating change in the event quarter, matching on the firm’s prior ESG rating, log market capitalization, MSCI industry classification, and lagged quarterly return. The matching is conducted separately for each of the four treatment groups: general upgrades, general downgrades, BB upgrades, and BB downgrades. Table 5 confirms adequate covariate balance across all four matched samples. Treated and control firms are nearly identical in mean size, prior ESG rating, industry composition, and lagged return, supporting the parallel trends assumption.

[INSERT TABLE 6 HERE]

Table 6 reports the PSM average treatment effects (ATE). For upgrades, the treated-minus-control difference in institutional ownership is 3.828% in the specification without firm- and time-fixed effects. When firm and time fixed effects are added to absorb remaining cross-sectional heterogeneity, the estimate falls to 0.373%. For downgrades, treated firms lose 2.613% of institutional ownership relative to controls in the specification without fixed effects, and 1.021% in the fixed-effects specification. The BB event subsamples produce larger-magnitude estimates for downgrades: -3.887% without fixed effects and -2.365% with fixed effects. BB upgrades show a positive treatment effect of 2.779% in the specification without fixed effects, but the estimate becomes small and insignificant once fixed effects are included.

[INSERT TABLE 7 HERE]

Table 7 presents the dynamic DiD estimates, which trace the ownership trajectory for treated and matched control firms from two quarters before to two quarters after the rating change. For upgrades, the pre-event coefficient at quarter $s-2$ is 0.083 and is not statistically significant, providing no evidence of a pre-existing trend. The treatment quarter coefficient is 0.101, which is also not statistically significant. The partial F-test comparing the treatment-quarter coefficient to the pre-event coefficient indicates no detectable ownership response at the time of the upgrade under this specification.

For downgrades, institutional ownership falls by 0.647% in the treatment quarter, with the partial F-test rejecting equality with pre-event ownership, with no reversal in $s+1$ and $s+2$. For BB downgrades, the treatment-quarter coefficient is -2.657

Taken together, the PSM and DiD results shift the interpretation of the TWFE evidence. The upgrade effect, which was significant in the fixed-effects specification, does not survive in the dynamic DiD, suggesting the TWFE estimate partly reflects pre-existing demand for firms that subsequently get upgraded rather than a response to the rating change itself. The downgrade effect is different. It is sharper, larger, and more persistent in the DiD than what TWFE suggested. Firms with a downgrade in ESG rating see institutional investors reduce holdings in a concentrated window around the rating change and do not recover in subsequent quarters. This effect is more pronounced for BB downgrades. This asymmetry between upgrades and downgrades is one of the more striking patterns in the data and points to a mandate-driven mechanism where the constraint binds more tightly on the downside than the upside.

This raises a natural question. Which investors are actually doing the selling? The aggregate 13F data hides variation across institution types, and whether the mechanism reflects mandate constraints or something else entirely. I explore this distinction in the next section.

6.4 Investor Type Decomposition

I decompose institutional ownership by investor class using the Bushee (1998, 2001) classification: Dedicated (DED), Quasi-Indexer (QIX), and Transient (TRA). DED and QIX investors hold longer-horizon, lower-turnover positions and are more likely to face ESG mandate constraints. TRA investors turn over holdings frequently and are less likely to be bound by ESG rating-based rules.

[INSERT TABLE 8 HERE]

Table 8 establishes the baseline result. DED holdings in AAA-rated firms are 4.86% higher than in CCC-rated firms; QIX holdings are 9.79% higher. TRA investors show the opposite: TRA holdings in CCC-rated firms are 2.01 percentage points higher than in AAA-rated firms. At the BB threshold, DED and QIX investors hold 1.64 and 4.45% more of above-threshold firms, respectively, while TRA shows a small negative difference of -0.35

[INSERT TABLE 9 HERE]

Table 9 shows that the aggregate upgrade effect is driven entirely by DED investors, who increase holdings by 1.442%. QIX investors show no response (0.028, not significant). TRA investors reduce holdings by 1.326%, moving in the opposite direction to DED. For downgrades, both DED and QIX reduce holdings, while TRA increases by 1.870%.

For BB-threshold events, the variations between investor classes become more salient. DED falls 2.665%, QIX falls 1.403%, and TRA rises 2.374%. At BB upgrades, the DED response is positive although not significant (0.782%); QIX shows an unexpected negative response of -1.970%, possibly reflecting rebalancing away from small-weight newly eligible names.

[INSERT TABLE 10 HERE]

Table 10 verifies these patterns in a pooled regression with investor-class interactions. The upgrade-DED interaction is 1.407%. The downgrade-DED interaction is -2.320%, and at BB downgrades falls further to -2.850%. The BB downgrade-QIX interaction is -1.588%, and the BB downgrade-TRA is 2.189%.

The net ownership change in the aggregate results reflects different institutional behavior. The DED and QIX investors reduce holdings while TRA increases on downgrades, with the reverse behavior on upgrades. This pattern is consistent with mandate-constrained investors rebalancing in response to rating changes, with TRA investors on the other side of the trade. The BB downgrade results, in which both DED and QIX respond significantly, suggest that the index eligibility threshold matters beyond the general rating signal.

6.5 ETF Channel and Parent-Index Conditioning

The decomposition shows that DED and QIX investors are driving the ownership response, particularly on the downgrade side. This raises the question of whether their rebalancing operates through the eligibility rules of ESG-focused indices (proxied by ETF constituent changes) that require funds to exit positions when a firm falls below the threshold. I next examine this channel. Further, major ESG index providers condition sub-index membership on concurrent membership in a parent broad-market index. Hence, some ETF addition and deletion events coincide with changes in broad-market index membership rather than reflecting ESG-specific rebalancing. To isolate the ESG component, I augment the baseline regression with indicator variables for ESG ETF additions and deletions and apply parent-index conditioning to separate ESG-driven ownership changes from those attributable to concurrent broad-market index events.

[INSERT TABLE 11 HERE]

Table 11 presents the index-based channel results, using ETFs as a proxy. Columns 1 through 3 include all ESG ETF additions and deletions without adjusting for concurrent

parent-index changes. The ETF deletion indicator is associated with a 1.361% decline in institutional ownership. The ETF addition indicator is positive at 0.143% but is not statistically significant. The upgrade coefficient in column 3 remains 0.278%, consistent with the earlier TWFE results. The ETF deletion result is economically large relative to the rating change effects, raising the question of whether it captures ESG-specific rebalancing or whether it inflates ESG ETF events with concurrent changes in parent broad-market index membership.

A key institutional feature of major ESG indices is that sub-index membership is conditioned on concurrent membership in a parent broad-market index. When a firm enters the parent broad-market index for the first time, it may be added to any ESG sub-index for which it qualifies, creating a bundled addition event. Similarly, when a firm exits the parent index, it is removed from all ESG sub-indices, creating bundled deletion events. These bundled events confound ESG-specific institutional demand with the well-documented ownership effects of broad-market index inclusion and exclusion. Of the 16,350 ESG ETF addition events associated with mapped parent indices in the sample, 420 events (2.57 percent) coincide with the firm's initial inclusion in the matched parent broad-market ETF in the same calendar month. The deletion sample shows a higher bundled rate, with 1,694 events (18.85 percent) coinciding with simultaneous parent-index exits.

Columns 4 through 6 present the parent-conditioned results, which exclude the bundled events from the sample. After conditioning, the ETF deletion coefficient falls to -0.280 and is no longer statistically significant. The addition coefficient is essentially unchanged at 0.161. The upgrade coefficient is robust to the conditioning and remains 0.278 throughout. These results show that the full-sample deletion effect is driven by firms that simultaneously exit an ESG sub-index and the underlying parent broad-market index, rather than by ESG-specific institutional rebalancing. Once this confound is removed, there is no statistically significant ownership effect at standalone ESG ETF deletion events. The persistence of the upgrade effect across all specifications further supports the interpretation that ESG rating upgrades generate a distinct positive institutional

ownership response beyond what can be assigned to index mechanics.

6.6 Coverage Initiation

Next, I examine a final channel through which MSCI ESG ratings affect institutional ownership: the initiation of coverage for previously unrated firms. Before receiving an MSCI rating, a firm is ineligible for ESG index inclusion regardless of its underlying characteristics. Coverage initiation removes this barrier and creates an extensive-margin shock to ESG-related institutional demand.

[INSERT TABLE 12 HERE]

Table 12 presents mean institutional ownership from two quarters before to two quarters after coverage initiation, stratified by the initial rating assigned. For the full sample, ownership rises from 62.60 percent at $s-2$ to 63.76 percent at $s+2$, an increase of 1.16 percent. The increase is larger for firms initiating at higher ratings: A-rated initiations show a gain of 2.96 percentage points, BBB initiations 2.54 percentage points, and AA initiations 1.13 percentage points. BB initiations show a smaller increase of 0.62 percentage, consistent with threshold uncertainty dampening accumulation by mandate-constrained investors. The one exception is CCC-rated initiations, where ownership declines by 1.25 percentage points following coverage. For these firms, MSCI coverage reveals a rating well below any mandate threshold, and the disclosure itself may prompt some institutional investors to reduce positions.

The coverage initiation results establish that entry into the MSCI universe generates an ownership response independent of subsequent rating changes. The response varies with the initial rating in a pattern consistent with the same rating-based mechanism documented in the earlier sections.

6.7 Value effect

I next examine whether the ownership changes documented above are accompanied by value effects. Tables 13 through 15 report cumulative abnormal returns (CARs) around ESG rating changes, ETF addition and deletion events, using a market-adjusted model. FF3-adjusted results are consistent and reported in the online appendix.

[INSERT TABLE 13 HERE]

Table 13 shows that ESG rating changes generate little abnormal return around the event day. For upgrades, the event-day CAR is -0.013% and not significant. For downgrades, the event-day CAR is 0.092%, positive and small. Upgrades show a modest decline of -0.146% over the (-5,-1) window, reflecting underperformance in the days before the rating change. The absence of a price reaction alongside the ownership shifts suggests investors rebalance without generating significant price pressure.

[INSERT TABLE 14 HERE]

Table 14 reports CARs around ESG ETF additions. Most windows are not significant. The exception is rating upgrades that precede ETF inclusion, where the 3-day CAR is 0.431%, consistent with a modest positive price effect around upgrades that subsequently trigger ETF additions.

[INSERT TABLE 15 HERE]

Table 15 reports CARs around ESG ETF deletions. CAR results around coverage initiation are reported in online appendix, where the full-sample 3-day CAR is 0.43%. A-rated initiations show the largest response at 1.05% over three days, consistent with the ownership increase documented in Table 12.

7 Conclusion

This paper investigates whether changes in MSCI ESG ratings influence institutional ownership and firm valuation using a U.S. firm-level data from 2013 to 2023. By combining MSCI ESG rating changes with 13F institutional holdings.

Overall, the results suggest that ESG upgrades are associated with a statistically significant increase in institutional ownership of 0.20 to 0.28 percent, and this result is robust across two-way fixed effects, propensity-score matching, and difference-in-differences specifications. Aggregate downgrades are not statistically significant in the baseline fixed-effects analysis. PSM-matched estimates, however, show ownership declines of 1.0 to 2.6 percent relative to matched controls, with larger and more persistent effects for downgrades that cross the BB threshold. The investor-type decomposition shows that the upgrade response is concentrated among Dedicated investors, while Transient investors move in the opposite direction. For downgrades, both Dedicated and Quasi-Indexed investors reduce holdings, while Transient investors again move oppositely.

The important finding of this paper concerns a confound in ESG ETF event studies. MSCI conditions ESG sub-index membership on concurrent membership in a parent broad-market index. As a result, 18.85 percent of deletion events in my sample coincide with exits from the matched parent index. The full-sample ETF deletion coefficient of -1.361 percent is significant. Once bundled events are excluded, this coefficient becomes insignificant. The rating-upgrade coefficient is unaffected by this conditioning. Any study that uses ESG-focused sub-index or ETFs deletions as a proxy for ESG-mandate selling should condition on parent broad-market index membership before drawing inference.

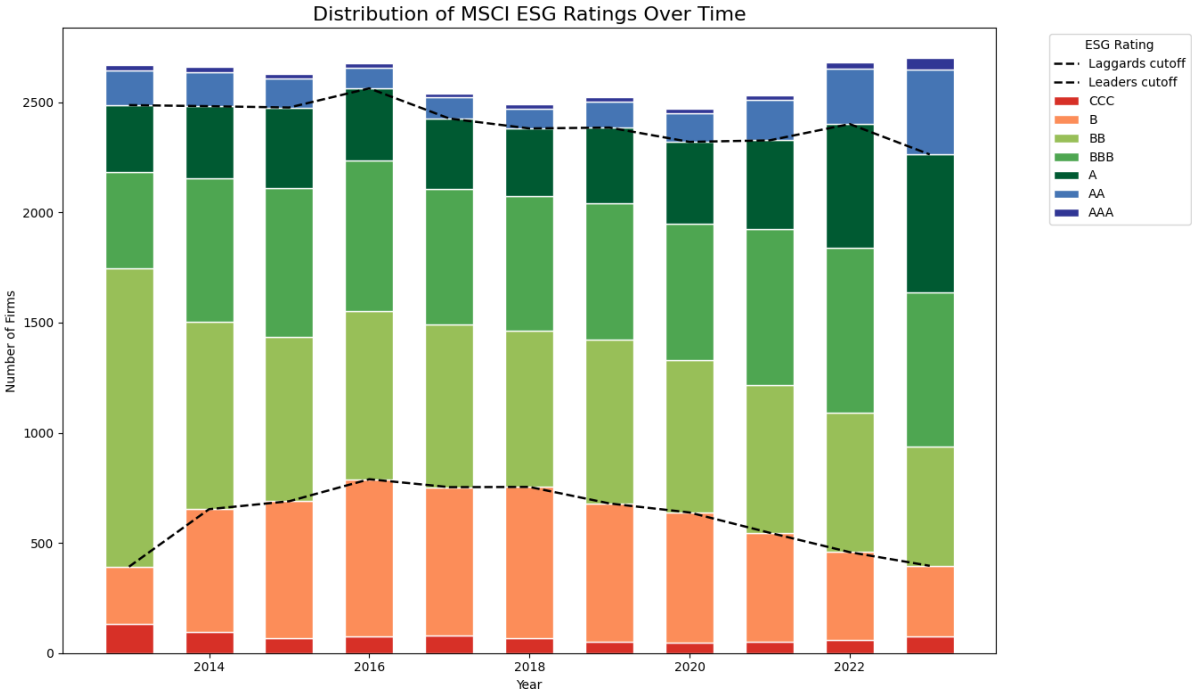
MSCI's 2012Q4 mass coverage initiation shows that the initial assignment of an ESG rating increases institutional ownership by 1.16 percent over the two quarters following initiation, with larger effects among firms receiving more favorable initial ratings. ESG rating changes generate little detectable short-window abnormal return, though ETF deletions are accompanied by significant negative CARs.

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Figure 1: MSCI ESG Ratings over Time



This figure reports the distribution of ESG ratings over time. The sample covers 28,571 firm-year observations from January 2013 to December 2023. Each company's Final Industry-Adjusted Score corresponds to a rating ranging from the highest (AAA) to the lowest (CCC). MSCI categorizes ESG ratings into three groups: *Leaders* (AAA and AA), *Average* (A, BBB, and BB), and *Laggards* (B and CCC).

Table 1: **Descriptive Statistics for Firm–Quarter-Level Variables**

Variable	p25	Mean	Median	p75	SD	N
13F Holding (%)	60.49	71.66	79.22	90.26	24.30	105,086
Change in 13F Holding	-1.52	0.06	0	1.79	5.71	100,656
Leverage	0.09	0.30	0.27	0.44	0.23	105,063
Log Market Cap	6.77	7.90	7.73	8.83	1.53	105,055
Market to Book	1.32	3.73	2.28	4.31	7.73	104,967
Profitability	-0.00	-0.00	0.01	0.02	0.04	105,062
Quarterly Return	-0.09	0.02	0.02	0.12	0.22	104,842
Rating Number	3.00	3.54	3.00	4.00	1.28	105,086

Note: This table shows descriptive statistics of the firm-quarter-level characteristics we use in our studies. Our sample comprises 4430 listed U.S. firms with available MSCI ESG rating data. Observations cover the period from January 2013 to December 2023. 13F Holding is the percentage of a company’s outstanding shares owned by 13F mutual funds. We obtain firm-level financial data from the CRSP-COMPUSTAT merged database. Leverage is a firm’s long-term debt plus debt in current liabilities, divided by total assets in percentage points. The log of firms’ market capital controls for the size of the firm. Book-to-market is the firms’ book value of equity divided by the market valuation. Profitability is income before extraordinary items over total assets. Quarterly return is an ex-dividend return for the past quarter. Rating number is MSCI’s assessment of how well companies manage ESG risks relative to industry peers on a scale from 1 to 7, with 1 being the worst (CCC) and 7 being the best (AAA).

Table 2: **ESG Rating Transition Matrix**

Current Rating	Previous Rating							Total
	AAA	AA	A	BBB	BB	B	CCC	
AAA	—	<i>79</i>	<i>12</i>	<i>3</i>	<i>2</i>	<i>1</i>	<i>0</i>	97
AA	24	—	<i>589</i>	<i>44</i>	<i>10</i>	<i>7</i>	<i>0</i>	679
A	6	285	—	<i>1,138</i>	<i>107</i>	<i>11</i>	<i>2</i>	1,549
BBB	4	35	583	—	<i>1,563</i>	<i>87</i>	<i>6</i>	2,278
BB	0	6	42	816	—	<i>1,093</i>	<i>26</i>	1,983
B	1	2	18	46	1,016	—	<i>165</i>	1,248
CCC	0	0	2	1	27	119	—	149
Total	40	407	1,246	2,048	2,725	1,318	199	7,983

Note: This table presents all observed quarter-to-quarter ESG rating transitions from 2013 to 2023. Rows indicate the current (post-transition) rating; columns indicate the prior-quarter rating. *Italicized green cells* represent upgrades (current rating higher than previous); *red cells* represent downgrades (current rating lower than previous); *gray cells* on the diagonal indicate no rating-category change. The sample includes 7,983 rating-change observations across 7 ESG rating categories. Of the 7,983 transitions shown, 1,113 are BB-crossing downgrades (BB or above to B or CCC) and 1,233 are BB-crossing upgrades (B or CCC to BB or above).

Table 3: **Univariate Analysis of Institutional Holdings (13F) by ESG Rating and BB Cutoff**

Panel A: Percentage of institutional holdings based on ESG rating								
ESG Rating	Count	Mean Holding	Median Holding	Leverage	Log Market Cap	Market-to-Book	Profitability	Quarterly Return
AAA	1084	76.94	81.03	0.29	9.70	5.97	0.02	0.024
AA	7023	78.64	83.77	0.31	8.77	5.46	0.01	0.021
A	16275	76.02	82.66	0.29	8.23	4.21	0.00	0.017
BBB	26322	73.52	80.53	0.29	7.93	3.76	-0.00	0.022
BB	30139	70.20	77.43	0.29	7.64	3.43	-0.00	0.024
B	21451	66.85	74.28	0.31	7.54	3.23	0.00	0.018
CCC	2792	61.89	71.44	0.34	8.25	2.54	0.01	0.019
Total	105086	71.66	79.22	0.30	7.90	3.73	0.00	0.021
ANOVA F-test: $F = 463.33$, $\text{Prob} > F = 0.0000$								
<i>Panel A Note:</i> This table reports statistics of institutional holdings (13F holdings) across ESG rating categories. The sample covers the period from January 2013 to December 2023. Mean holdings are reported along with medians. 13F holdings are reported in percentages. ANOVA F-test indicates significant differences in mean holdings across ESG ratings.								
Panel B: Percentage of institutional holdings based on BB cutoff								
BB Cutoff	Count	Mean Holding	Median Holding	Leverage	Log Market Cap	Market-to-Book	Profitability	Quarterly Return
Above	80843	73.28	80.43	0.29	7.98	3.91	-0.00	0.022
Below	24243	66.28	73.93	0.31	7.63	3.15	0.00	0.018
Total	105086	71.66	79.22	0.30	7.90	3.73	0.00	0.021
Difference (Above – Below): 6.997***								
<i>Panel B Note:</i> This panel categorizes firms according to their average 13F holdings relative to the BB rating cutoff. “Above” includes firms with ESG ratings of BB or higher (BB, BBB, A, AA, AAA), while “Below” includes firms with ESG ratings lower than BB (B, CCC). Observations cover January 2013 to December 2023.								
Panel C: Percentage of institutional holdings for BB cutoff rating changes								
BB Cutoff	Count	Mean Holding	Median Holding	Leverage	Log Market Cap	Market-to-Book	Profitability	Quarterly Return
Upgrade	1233	70.38	78.81	0.31	7.92	3.42	0.00	0.04
Downgrade	1113	67.09	74.64	0.29	7.45	8.41	-0.00	0.30
Difference (Upgrade – Downgrade): 3.28***								
<i>Panel C Note:</i> This panel presents firms experiencing rating changes around the BB cutoff. “Upgrade” refers to firms that improved their rating across BB, while “Downgrade” refers to firms that fell below BB. Observations cover January 2013 to December 2023.								

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Table 4: **ESG Ratings and Institutional Ownership**

VARIABLES	(1)	(2)	(3)
	13F Holding (%)		
Upgrade Indicator	0.203*		0.278**
	(1.690)		(2.063)
Downgrade Indicator	-0.142		0.051
	(0.806)		(0.245)
BB Upgrade Indicator		-0.055	-0.303
		(-0.207)	(-1.021)
BB Downgrade Indicator		-0.491	-0.526
		(-1.510)	(-1.365)
Constant	33.60***	33.61***	33.61***
	(15.06)	(15.07)	(15.07)
Observations	104,895	104,895	104,895
R-squared	0.882	0.882	0.882
Controls	YES	YES	YES
Firm FE	YES	YES	YES
Time FE	YES	YES	YES

Notes: The table reports regression results of the type presented in Equation $y_{it} = \alpha + \delta \text{Upgrade} + \theta \text{Downgrade} + \rho \text{BBUpgrade} + \sigma \text{BBDowngrade} + \mu_i + \theta_t + \varepsilon_{it}$, where y_{it} is 13F Holding (percentage of a company's outstanding shares owned by 13F funds). Upgrade and Downgrade are indicator variables equal to 1 if the firm experienced an ESG rating upgrade or downgrade, respectively. Rating changes around the BB threshold determine a firm's eligibility for major ESG indices. BB Upgrade and BB Downgrade are indicator variables equal to 1 if the rating change crosses the BB eligibility threshold. Specification (1) includes the general upgrade and downgrade indicators only. Specification (2) includes only the BB-threshold indicators. Specification (3) includes all four indicators jointly. t -statistics based on standard errors clustered at the event level are shown in parentheses. Asterisks denote statistical significance (* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$).

Table 5: **Descriptive Statistics for Propensity Score Matching (PSM) Samples**

Variable	Treated Sample			Control Sample		
	N	Mean	SD	N	Mean	SD
Treatment: Upgrades						
Size	4,403	8.21	1.56	4,269	8.22	1.65
ESG rating before upgrade	4,403	3.19	1.10	4,269	3.19	1.19
MSCI Indus	4,403	44.81	25.76	4,269	44.04	25.88
Quarterly Return	4,403	0.04	0.26	4,269	0.03	0.26
Treatment: Downgrades						
Size	2,624	7.77	1.55	2,543	7.81	1.55
ESG rating before downgrade	2,624	4.03	1.11	2,543	4.03	1.17
MSCI Indus	2,624	44.00	25.67	2,543	43.85	25.78
Quarterly Return	2,624	0.05	0.34	2,534	0.05	0.41
Treatment: BB Upgrades						
Size	1,121	7.92	1.49	1,186	7.60	1.41
ESG rating before downgrade	1,121	1.97	0.16	1,186	1.92	0.28
MSCI Indus	1,121	44.30	26.01	1,186	44.76	25.64
Quarterly Return	1,121	0.03	0.22	1,186	0.01	0.23
Treatment: BB Downgrades						
Size	1,112	7.45	1.35	1,103	7.35	1.23
ESG rating before downgrade	1,112	3.09	0.37	1,103	2.95	1.01
MSCI Indus	1,112	44.19	26.03	1,103	43.12	25.94
Quarterly Return	1,112	0.05	0.23	1,103	0.03	0.21

Notes: This table presents summary statistics for key firm-level variables used in the propensity score matching (PSM) procedure, comparing treated and control groups for ESG upgrades, ESG downgrades, BB Upgrades and BB Downgrades. Variables include firm size (log of market capitalization), ESG rating prior to the event, MSCI industry classification, and quarterly stock return. The statistics confirm strong balance across observable characteristics between treated and control samples, supporting the validity of the matching strategy.

Table 6: **PSM: Average Treatment Effect – 13F Holdings around ESG Rating Changes**

Dependent Variable	13F Holding							
	Upgrade (1)	Upgrade (2)	Downgrade (3)	Downgrade (4)	BB Upgrade (5)	BB Upgrade (6)	BB Down- grade (7)	BB Down- grade (8)
Rating Change	3.828*** (7.60)	0.373* (1.66)	-2.613*** (-4.02)	-1.021*** (-2.77)				
BB Rating Change					2.779** (2.65)	-0.170 (-0.29)	-3.887*** (-3.64)	-2.365*** (-2.71)
Observations	8,672	7,778	5,158	3,865	2,407	1,694	2,215	1,157
Adj. R-squared	0.035	0.879	0.050	0.854	0.031	0.888	0.034	0.872
Controls	YES	YES	YES	YES	YES	YES	YES	YES
Firm FE	NO	YES	NO	YES	NO	YES	NO	YES
Time FE	NO	YES	NO	YES	NO	YES	NO	YES

Notes: This table reports regression results analyzing the impact of ESG rating changes on institutional ownership, measured by 13F holdings. The sample includes firms experiencing four types of rating changes: Upgrade, Downgrade, BB Upgrade, and BB Downgrade. For each treated firm, a matched control group is constructed via propensity score matching, based on prior rating, firm size, MSCI industry classification, and lagged quarterly returns, with matching starting from the quarter of the rating change. The coefficients represent the difference in 13F holdings between treated firms and matched controls in the quarter of the rating event. The models include indicator variables for rating changes (main effects) and interactions between capturing rating changes around the BB cutoff threshold. Columns vary by fixed effects: firm and time fixed effects are included in some specifications to control for unobserved heterogeneity. t-statistics are reported in parentheses. Statistical significance is denoted as follows: ***p < 0.01, **p < 0.05, *p < 0.10 (two-tailed tests).

Table 7: **Dynamic model– 13F Holding around ESG Rating Changes**

Dependent Variable	13F Holding			
	Upgrade	Downgrade	BB Upgrade	BB Downgrade
Sample	(1)	(2)	(3)	(4)
Rating Change d[s-2]	0.083 (0.74)	0.400** (2.67)		
Rating Change	0.101 (1.02)	-0.647*** (-4.36)		
Rating Change d[s+1]	-0.104 (-0.91)	0.237 (1.55)		
Rating Change d[s+2]	-0.108 (-0.86)	0.235 (1.44)		
BB Rating Change d[s-2]			0.480 (0.50)	1.725 (0.59)
BB Rating Change			-0.059 (-0.08)	-2.657** (-2.03)
BB Rating Change d[s+1]			-0.524 (-0.32)	0.152 (0.08)
BB Rating Change d[s+2]			-1.333 (-0.47)	-3.958 (-1.55)
Observations	35,347	22,180	1,362	947
Adj. R-squared	0.903	0.900	0.877	0.876
Controls	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES
Time FE	YES	YES	YES	YES
<i>p -values for partial F -tests comparing coefficients</i>				
d[s] vs d[s-2]	0.447	0.000	0.877	0.094
d[s+1] vs d[s-2]	0.392	0.019	0.841	0.840
d[s+2] vs d[s-2]	0.421	0.021	0.797	0.246

Notes: This table presents difference-in-differences analyses of 13F holdings around rating changes. Columns 1 and 2 examine the Upgrades and Downgrades in ESG ratings and models 3 and 4 control for the rating changes around the BB cutoff point. For target firms that experience rating changes (“Rating Change”; all columns), we select untargeted control firms (based on size, previous ESG rating, MSCI Industry, and past quarter returns) by propensity score matching in the quarter that the rating changes. For target firms that experience a rating change around the BB cutoff point (columns 3 and 4), we select untargeted control firms by propensity score matching on the no rating change firms. The tabulated coefficients measure how the difference between treatment firm 13F holding and control firm 13F holding has changed from the quarter before treatment/placebo (the omitted base quarter) to quarter $s \pm k$. All columns include indicator variables for the main effect of the rating change (BB rating change). All columns include a set of 4 $d[s \pm k]$ indicator variables, one for each quarter relative to the rating change quarter (for treatment firms) and matched placebo quarter (for control firms). In addition, columns 3 and 4 include a set of 4 $d[s \pm k]$ indicator variables, one for each quarter relative to the BB cutoff rating change and matched placebo quarter. At the bottom of the table, p -values of partial F -tests indicate the statistical significance of difference-in-differences estimations of 13F holding around rating changes using quarter $s - 1$ as the base quarter. Coefficients and standard errors are estimated using OLS. t -statistics appear in parentheses. ***, **, * denote statistical significance at the 1%, 5%, and 10% levels, respectively (two-tailed).

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Table 8: Mean Holdings by ESG Rating and BB Cutoff, by Investor Class

Panel A: Holdings by ESG Rating and Investor Class				
	Dedicated	Quasi-indexer	Transient	Unknown
AAA	17.27 (1,045)	40.93 (1,089)	7.27 (1,088)	7.67 (1,820)
AA	18.07 (6,724)	39.77 (7,031)	8.29 (6,996)	9.29 (10,668)
A	17.48 (15,319)	38.00 (16,184)	9.04 (16,110)	9.61 (24,077)
BBB	16.24 (24,474)	35.82 (26,148)	9.82 (26,056)	9.31 (38,779)
BB	14.69 (26,888)	34.27 (29,741)	10.00 (29,466)	8.95 (44,067)
B	14.72 (19,311)	31.72 (21,243)	10.00 (21,143)	8.63 (30,782)
CCC	12.42 (2,338)	31.13 (2,729)	9.28 (2,713)	7.32 (4,260)
Total	15.75 (96,099)	35.08 (104,165)	9.64 (103,572)	9.04 (154,453)
Difference (AAA–CCC)	4.86***	9.79***	-2.01***	0.35*
Panel B: Holdings by BB Cutoff and Investor Class				
	Dedicated	Quasi-indexer	Transient	Unknown
Above	16.12 (74,450)	36.10 (80,193)	9.56 (79,716)	9.21 (119,411)
Below	14.47 (21,649)	31.65 (23,972)	9.92 (23,856)	8.47 (35,042)
Total	15.75 (96,099)	35.08 (104,165)	9.64 (103,572)	9.04 (154,453)
Difference	1.64***	4.45***	-0.35***	0.75***

Notes: Panel A reports mean holdings (in percentage terms) by ESG rating category and investor class. Panel B reports mean holdings by BB cutoff and investor class. Investor classes include Dedicated (DED), Quasi-indexer (QIX), Transient (TRA), and unmatched investors (Unknown). For Panel A, the bottom row reports the difference in mean holdings between AAA- and CCC-rated firms. For Panel B, the difference in mean holdings between above BB_cutoff and below BB_cutoff tested using a t-test; sample sizes are reported in parentheses.

Table 9: **Marginal Effects of Rating Changes and BB Threshold by Investor Class**

	Any Rating Change			BB Threshold		
	DED	QIX	TRA	DED	QIX	TRA
Panel A: Upgrade						
Indicator	1.442***	0.028	-1.326***	0.782	-1.970***	0.351
Std. Err.	(0.258)	(0.297)	(0.176)	(0.515)	(0.579)	(0.351)
Panel B: Downgrade						
Indicator	-2.148***	-0.705*	1.870***	-2.665***	-1.403**	2.374***
Std. Err.	(0.313)	(0.362)	(0.224)	(0.485)	(0.594)	(0.357)
Observations	458,286	458,286	458,286	458,286	458,286	458,286
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table reports marginal effects from regressions of institutional holdings on indicators for any rating change (columns 2–4) and crossing the BB threshold (columns 5–7). Panel A reports upgrades; Panel B reports downgrades. Investor classes include dedicated (DED), quasi-indexers (QIX), and transient (TRA). Standard errors are reported below the coefficients in parentheses and are clustered at the firm level. Observations correspond to the full sample, and firm and quarter fixed effects are included. *Significance levels:* *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 10: **Effect of ESG rating changes on holding by Investor Class**
Interaction for class in regression

	Any Rating Change	BB Threshold
Up Indicator	0.034 (0.158)	0.490 (0.326)
Up $\times$ DED	1.407*** (0.322)	0.292 (0.630)
Up $\times$ QIX	-0.007 (0.383)	-2.460*** (0.757)
Up $\times$ TRA	-1.360*** (0.237)	-0.139 (0.478)
Down Indicator	0.172 (0.180)	0.185 (0.270)
Down $\times$ DED	-2.320*** (0.372)	-2.850*** (0.565)
Down $\times$ QIX	-0.877* (0.466)	-1.588** (0.735)
Down $\times$ TRA	1.698*** (0.296)	2.189*** (0.438)
Observations	458,286	458,286
Firm Fixed Effects	Yes	Yes
Quarter Fixed Effects	Yes	Yes

Notes: This table reports coefficient estimates from regressions of institutional holdings on indicators for any upward or downward rating change (column 1) and crossing the BB rating threshold (column 2). The specification includes interactions between the indicators and investor class indicator variables, with the omitted category corresponding to unmatched investors (Unknown). Firm and quarter fixed effects are included, and standard errors are clustered at the firm level. Coefficients on interaction terms represent differential effects relative to the omitted investor class. Standard errors are reported in parentheses.

Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 11: **Parent-Index Conditioning: Full Sample versus Clean Sample**

VARIABLES	<i>Full Sample</i>			<i>Parent-Conditioned Sample</i>		
	(1)	(2)	(3) 13F Holding (%)	(4)	(5)	(6)
Upgrade Indicator	0.202* (1.687)		0.278** (2.603)	0.203* (1.681)		0.278** (2.598)
Downgrade Indicator	-0.143 (-0.809)		0.051 (0.244)	-0.141 (-0.797)		0.053 (0.251)
BB Upgrade Indicator		-0.056 (-0.213)	-0.305 (-1.027)		-0.058 (-0.221)	-0.307 (-1.031)
BB Downgrade Indicator		-0.492 (-1.514)	-0.527 (-1.368)		-0.491 (-1.511)	-0.526 (-1.365)
Added	0.143 (0.819)	0.145 (0.830)	0.145 (0.829)	0.161 (0.942)	0.163 (0.951)	0.163 (0.950)
Removed	-1.361** (-2.034)	-1.361** (-2.034)	-1.361** (-2.035)	-0.280 (-0.480)	-0.279 (-0.477)	-0.280 (-0.479)
Constant	33.60*** (15.06)	33.60*** (15.07)	33.61*** (15.07)	33.58*** (15.03)	33.58*** (15.04)	33.59*** (15.05)
Observations	104,895	104,895	104,895	104,895	104,895	104,895
R-squared	0.882	0.882	0.882	0.882	0.882	0.882
Controls	YES	YES	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES	YES	YES
Time FE	YES	YES	YES	YES	YES	YES

Notes: The table reports two sets of regression results from the specification $y_{it} = \alpha + \delta \text{Upgrade} + \theta \text{Downgrade} + \rho \text{BBUpgrade} + \sigma \text{BBDowngrade} + \text{Added} + \text{Removed} + \mu_i + \theta_t + \varepsilon_{it}$, where y_{it} is the percentage of a company's outstanding shares owned by 13F funds. The left three columns (Full Sample) use all 16,350 ESG ETF addition and 8,988 deletion events. The right three columns (Parent-Conditioned Sample) exclude the 420 additions (2.57%) that coincide with the firm's first entry into the matched parent broad-market ETF in the same calendar month, isolating ESG-specific demand from concurrent broad-market index inclusion effects. Upgrade and Downgrade are indicator variables equal to 1 if the firm experienced an ESG rating upgrade or downgrade, respectively. BB Upgrade and BB Downgrade are indicator variables equal to 1 if the rating change crosses the BB eligibility threshold for major ESG indices. Added and Removed are indicator variables equal to 1 if the firm is added to or removed from an ESG-focused ETF. Specifications (1) and (4) include general rating-change indicators only; Specifications (2) and (5) include BB-threshold indicators only; Specifications (3) and (6) include all four indicators jointly. t -statistics based on standard errors clustered at the event level are shown in parentheses. Asterisks denote statistical significance (* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$).

Table 12: Institutional Ownership Around MSCI Coverage Initiation by Initial MSCI Rating

Panel A: Full Sample, BB, and BBB										
Time Relative to Initiation	Full Sample			BB			BBB			
	N	Mean	Median	N	Mean	Median	N	Mean	Median	
$s - 2$	1594	62.60	70.15	1026	62.72	69.72	197	60.86	69.91	
$s - 1$	1596	62.51	69.96	1029	62.45	69.01	201	59.43	68.62	
s	1602	62.55	70.14	1030	62.22	69.19	201	61.26	68.62	
$s + 1$	1574	62.80	70.48	1011	62.39	69.22	201	61.83	69.18	
$s + 2$	1556	63.76	71.56	997	63.34	70.41	200	63.40	72.07	
Diff										
$(s + 2 - s - 2)$		1.16			0.62					2.54
Panel B: AA, A, B, and CCC										
Time Relative to Initiation	AA			A			B			CCC
	N	Mean	Median	N	Mean	Median	N	Mean	Median	
$s - 2$	94	70.47	79.20	160	61.20	70.17	98	60.17	66.98	74.45
$s - 1$	91	72.94	79.62	159	62.41	72.54	97	60.63	67.49	75.15
s	94	70.17	78.93	160	62.41	72.79	98	62.01	69.41	75.69
$s + 1$	90	70.16	78.01	157	63.07	73.00	96	62.62	71.22	73.76
$s + 2$	90	71.61	78.35	154	64.16	74.37	97	61.97	70.54	71.73
Diff										
$(s + 2 - s - 2)$		1.13			2.96			1.79		-1.25

Notes: This table reports summary statistics for institutional ownership around MSCI coverage initiation. Initial MSCI ratings are defined as the rating observed in the initiation quarter (2012Q4) and are held constant across all firm-quarters in the event window. Differences are based on t -tests comparing institutional ownership in quarters $s - 2$ and $s + 2$.

Table 13: Descriptive Statistics for CAR around Rating Changes

Variable	p25	Mean	Median	p75	SD	N
Panel A: Market-Adjusted CAR						
ESG rating Upgrades						
CAR (0,0) day	-0.8900	-0.0133	-0.0135	0.9030	2.0558	4,895
CAR (0,2) days	-1.7128	0.0186	-0.0081	1.7513	3.7086	4,895
CAR (0,4) days	-2.2278	0.0865	-0.0419	2.2905	4.7234	4,895
CAR (-1,-1) day	-0.9417	0.0265	0.0185	0.9684	2.0521	4,895
CAR (-3,-1) days	-1.6989	-0.0189	0.0195	1.7083	3.6312	4,895
CAR (-5,-1) days	-2.3039	-0.1458**	-0.0252	2.2780	4.8493	4,894
ESG rating Downgrades						
CAR (0,0) day	-0.8910	0.0924**	0.0061	1.0221	2.0258	2,995
CAR (0,2) days	-1.6282	0.1005	0.0431	1.8282	3.7223	2,995
CAR (0,4) days	-2.3301	0.0083	-0.0748	2.2646	4.8767	2,995
CAR (-1,-1) day	-0.8973	0.0624*	0.0003	0.9589	2.0564	2,995
CAR (-3,-1) days	-1.6013	0.1700***	0.0083	1.7933	3.5718	2,995
CAR (-5,-1) days	-2.1776	0.0433	-0.0790	2.2793	4.8499	2,995
ESG rating BB Upgrades						
CAR (0,0) day	-0.9864	-0.0354	0.0221	0.9413	2.2638	1,289
CAR (0,2) days	-1.7565	0.0817	0.1368	1.8299	3.9533	1,289
CAR (0,4) days	-2.4020	0.2260	0.0376	2.3815	5.0843	1,289
CAR (-1,-1) day	-0.8816	0.2157***	0.0762	1.1039	2.3115	1,289
CAR (-3,-1) days	-1.5814	0.2180**	0.0434	1.9669	3.9270	1,289
CAR (-5,-1) days	-2.1764	0.1654	0.0577	2.5515	5.0987	1,289
ESG rating BB Downgrades						
CAR (0,0) day	-0.8382	0.1231**	0.0665	1.1332	1.9763	1,175
CAR (0,2) days	-1.4842	0.1843*	0.0440	1.8799	3.7907	1,175
CAR (0,4) days	-2.2276	0.0109	-0.1023	2.1845	4.8437	1,175
CAR (-1,-1) day	-0.8662	0.0960	0.0498	1.0618	2.0570	1,175
CAR (-3,-1) days	-1.5618	0.2799**	0.0115	1.8701	3.7882	1,175
CAR (-5,-1) days	-2.3286	0.1420	-0.0977	2.2960	4.9020	1,175

Note: This table reports descriptive statistics for firm-level cumulative abnormal returns, CAR, in percentage points using the market-adjusted model around ESG rating changes. CAR variables are winsorized at the 1st and 99th percentiles. Post-event windows are measured over days (0,0), (0,2), and (0,4). Pre-event windows are measured over days (-1,-1), (-3,-1), and (-5,-1). Observations cover the period from January 2013 to December 2023. Upgrades and downgrades refer to ESG rating increases and decreases, respectively. BB upgrades and BB downgrades refer to rating changes around the BB cutoff, where the rating adjustment moves the firm above or below the BB threshold. Significance stars are based on two-sided one-sample t-tests of whether the mean CAR equals zero:

* p<0.10, ** p<0.05, and *** p<0.01.

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Table 14: **Cumulative Abnormal Returns (CARs) Around ESG Rating Upgrades and ETF Additions**

Days	CAR at Upgrade (Full)	CAR at Added (Full)	CAR at Upgrade (Same Month)	CAR at Added (Same Month)	CAR at Upgrade (≤ 4 mo)	CAR at Added (≤ 4 mo)	CAR at Upgrade (Before)	CAR at Added (Before)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: CAR based on Market Model								
(0,0)	0.0865	0.1096	-0.1938	-0.2092	-0.0478	0.2265	0.2396	0.1236
(-1,0)	0.0997	0.1085	0.0131	0.0083	0.0540	0.0085	0.1494	0.1941
(0,2)	0.1727	0.1657	0.0753	-0.5174	-0.2193	0.3770	0.4309**	0.2193
(-3,2)	0.2011	-0.0006	0.3233	-0.3282	-0.1348	-0.1105	0.3685	0.1505
(0,4)	0.1105	0.2079	0.2708	-0.7870*	-0.3653	0.2680	0.3509	0.4328
(-5,4)	-0.0231	0.0020	0.4625	-0.2182	-0.6159	-0.3536	0.2016	0.2707
Panel B: CAR based on FF 3 factor								
(0,0)	-0.0189	0.1366*	-0.2997*	-0.0126	-0.1323	0.3006**	0.1236	0.0794
(-1,0)	-0.0248	0.1011	-0.1473	0.2579	-0.0769	0.1146	0.0390	0.0519
(0,2)	-0.1024	0.0680	-0.3244	-0.4063*	-0.4557*	0.2582	0.1677	0.0809
(-3,2)	-0.0659	-0.0560	-0.0681	-0.1654	-0.3185	-0.0196	0.0856	-0.0486
(0,4)	0.0062	0.1643	0.2941	-0.7911**	-0.5019*	0.3766	0.2330	0.2910
(-5,4)	-0.1714	0.0442	0.4566	-0.1057	-0.7519*	-0.0585	0.0081	0.1440
N	1,091	1,091	154	154	349	349	588	588

Notes: This table reports average cumulative abnormal returns (CARs) over various event windows centered on ESG rating upgrades and ETF additions. Panel A presents CARs estimated using the market model, while Panel B reports CARs estimated using the Fama–French three-factor model. Each column reflects a different timing condition. “Full” indicates the entire sample; “Same Month” refers to firms added to ESG ETFs in the same month as the rating upgrade; “ ≤ 4 mo” includes additions within four months after the rating change; and “Before” includes ETF additions that occurred prior to the ESG rating upgrade. CARs are expressed in percentage points and tested against zero using one-sample t-tests. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 15: **Cumulative Abnormal Returns (CARs) Around ESG Rating Downgrades and ETF Removals**

Days	CAR at Down- grade (Full)	CAR at Removed (Full)	CAR at Down- grade (Same Month)	CAR at Removed (Same Month)	CAR at Down- grade (≤ 4 mo)	CAR at Removed (≤ 4 mo)	CAR at Down- grade (Before)	CAR at Removed (Before)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: CAR based on Market Model								
(0,0)	-0.2387	-0.3054	-0.3165	-0.6929*	-0.0787	-0.5585	-0.3669	0.1603
(-1,0)	-0.0935	-0.4097	-0.7631	-0.7378	0.6359*	-0.7287	-0.5165	0.0946
(0,2)	-0.5089**	-0.3927	-0.5694	-0.9458*	-0.4531	-0.6917	-0.5364	0.2064
(-3,2)	-0.3924	-0.6389	-1.3230	-1.2697*	0.6158	-0.5499	-0.9742*	-0.4079
(0,4)	-0.9650***	-0.2535	-0.5271	-1.3139**	-0.8630*	-0.4870	-1.2975***	0.5376
(-5,4)	-0.7716*	-0.3233	-1.1427	-1.9308**	0.2503	-0.2024	-1.6558***	0.3764
Panel B: CAR based on FF 3 factor								
(0,0)	-0.0630	-0.3203**	-0.0109	-0.6412**	0.0971	-0.2157	-0.2582*	-0.2653
(-1,0)	0.0050	-0.3835**	-0.4594	-0.7942*	0.6891**	-0.3202	-0.4757**	-0.2387
(0,2)	-0.4182	-0.6663**	-1.3373***	-1.3374***	0.0820	-0.9780*	-0.2154	0.0069
(-3,2)	-0.7989**	-0.9786**	-2.0526***	-1.7301***	0.3000	-1.1431*	-0.7409*	-0.4190
(0,4)	-0.6056**	-0.6246*	-0.4160	-1.6323***	-0.3393	-0.7878	-0.9832**	0.0654
(-5,4)	-0.7082*	-0.9747*	-1.3529	-2.5993***	0.4426	-0.9248	-1.5873***	-0.1914
N	444	444	89	89	182	182	173	173

Notes: This table reports average cumulative abnormal returns (CARs) over various event windows centered on ESG rating downgrades and ETF removals. Panel A presents CARs estimated using the market model, while Panel B reports CARs estimated using the Fama–French three-factor model. Each column reflects a different timing condition. “Full” indicates the entire sample; “Same Month” refers to firms removed from ESG ETFs in the same month as the rating downgrade; “ ≤ 4 mo” includes removals within four months after the rating change; and “Before” includes ETF removals that occurred prior to the ESG rating downgrade. CARs are expressed in percentage points and tested against zero using one-sample t-tests. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.